A NEGATIVE ANSWER TO PROBLEM 8.8 OF GALETTO, MONTAÑO, AND WELLNER

Abstract. Let $K = \mathrm{VR}(Q_6; 4)$ and $G = \mathrm{Aut}(Q_6)$. We prove that, over every field $\Bbbk$ of characteristic zero, and conditional on the Betti numbers of K computed in [1, Appendix C.3],

$$\dim_{\Bbbk} H_7(K;\Bbbk)^G + \dim_{\Bbbk} H_{15}(K;\Bbbk)^G = 3.$$

Thus the trivial representation occurs with multiplicity at least two in one of these two homology groups. This gives a negative answer to Problem 8.8 of Galetto, Montaño, and Wellner. The proof is an exact Lefschetz computation on the 65 conjugacy classes of the hyperoctahedral group.

1. The counterexample

Let Q_n be the n-dimensional hypercube graph, and identify it throughout with its vertex set $\{0,1\}^n$ carrying the Hamming metric d, which is the graph metric of Q_n. In particular $\mathrm{Aut}(Q_n)$ is simultaneously the automorphism group of the graph and the isometry group of the metric space, and the Vietoris–Rips complex below is taken on that vertex set:

$$X^{n,r} = \mathrm{VR}(Q_n; r) = \{\sigma \subseteq Q_n : d(x,y) \le r \text{ for all } x, y \in \sigma\}.$$

The hyperoctahedral group

$$\mathfrak{H}_n = \mathrm{Aut}(Q_n) \cong \mathbb{F}_2^n \rtimes \mathfrak{S}_n$$

acts simplicially on $X^{n,r}$. Galetto, Montaño, and Wellner proved multiplicity-freeness of the nonzero homology representations at scales $r \le 3$ and $r = n-1$, and asked whether $H_i(X^{n,r})$ is multiplicity-free for all n, r, i [1, Problem 8.8]. The smallest pair with $4 \le r \le n-2$ is $(n,r) = (6,4)$. They also computed

$$\widetilde{H}_i(X^{6,4};\mathbb{Q}) \cong \begin{cases} \mathbb{Q}^{239}, & i = 7, \\ \mathbb{Q}^{14}, & i = 15, \\ 0, & i \notin \{7, 15\}. \end{cases} \tag{1}$$

See [1, Appendix C.3]. Equation (1) is a machine computation of theirs which we do not reproduce; it is the only computational input taken as given here, and Theorem 1 is conditional on it, both for the vanishing of $\widetilde{H}_i(X^{6,4};\mathbb{Q})$ outside $\{7, 15\}$ and for the parity of the two surviving degrees.

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Theorem 1. *Let $K = X^{6,4}$ and $G = \mathfrak{H}_6$. For every field $\Bbbk$ of characteristic zero,*

$$\dim_{\Bbbk} H_7(K;\Bbbk)^G + \dim_{\Bbbk} H_{15}(K;\Bbbk)^G = 3.$$

Consequently, at least one of the G-representations $H_7(K;\Bbbk)$ and $H_{15}(K;\Bbbk)$ is not multiplicity-free.

The argument identifies the counterexample at $(n, r) = (6, 4)$ but does not determine which of the two degrees contains the repeated trivial constituent.

2. A fixed-orbit Lefschetz formula

For $g \in G$, define

$$\Psi(g) = \sum_{i\geq 0} (-1)^{i+1} \operatorname{tr}\Big(g \mid \widetilde{H}_i(K;\mathbb{Q})\Big).$$

Let $\mathcal{O}_g$ be the set of g-orbits on Q_6 with internal Hamming diameter at most 4. Define a graph F_g on $\mathcal{O}_g$ by joining O, O' when

$$\operatorname{diam}(O \cup O') > 4.$$

For a graph F, write

$$I(F;t) = \sum_{S\in \operatorname{Ind}(F)} t^{|S|}$$

for its independence polynomial.

Lemma 2. *For every $g \in G$,*

$$\Psi(g) = I(F_g; -1).$$

Proof. A simplex fixed setwise by g is a union of vertex orbits. Such a union is a simplex of K exactly when the selected orbits form an independent set of F_g.

Suppose a fixed simplex σ has s vertices and is the union of c vertex orbits. The induced permutation of its vertices has c cycles, hence orientation sign $(-1)^{s-c}$. Since σ lies in chain degree $s-1$, its contribution to the reduced Lefschetz number is

$$(-1)^{s-1}(-1)^{s-c} = (-1)^{c+1}.$$

The empty simplex in degree -1 contributes -1, which is the same formula for $c = 0$. The Hopf trace formula, applied to the augmented chain complex (indexed by $i \geq -1$) and its reduced homology, therefore gives

$$\sum_{i\geq -1} (-1)^i \operatorname{tr}\Big(g \mid \widetilde{H}_i(K;\mathbb{Q})\Big) = -\sum_{S\in\operatorname{Ind}(F_g)} (-1)^{|S|}.$$

Here $\widetilde{H}_{-1}(K) = 0$ because K is nonempty, so the left-hand side reduces to the sum over $i \geq 0$, which equals $-\Psi(g)$. Multiplying by -1 proves the claim. $\square$

3. Exact class computation

Write $g = (a, \pi) \in \mathbb{F}_2^6 \rtimes \mathfrak{S}_6$, acting by $x \mapsto a + \pi x$. For each cycle C of π, the parity $\sum_{j \in C} a_j \in \mathbb{F}_2$ is invariant under conjugacy. These parities, together with the cycle lengths, form a complete invariant: conjugacy classes are indexed by bipartitions $(\alpha; \beta)$ of 6, where α records the even-parity cycles and β the odd-parity cycles [1, Section 5.2].

If $m_j(\lambda)$ is the multiplicity of j in a partition λ, set

$$z_\lambda = \prod_{j \geq 1} j^{m_j(\lambda)} m_j(\lambda)!.$$

A signed cycle of length j contributes a factor $2j$ to the centralizer, and signed cycles of the same length and parity may be permuted. Hence

$$|C_{\alpha,\beta}| = \frac{2^6 6!}{2^{\ell(\alpha)+\ell(\beta)} z_\alpha z_\beta}. \tag{2}$$

There are 65 bipartitions of 6.

For each type $(\alpha; \beta)$, choose disjoint cycles on consecutive coordinate blocks, take $a = 0$ on every α-cycle, and put a single 1 in a on every β-cycle. Lemma 2 then reduces the class value to an exact graph computation. If $J(F) = I(F; -1)$ and $N[v]$ is the closed neighborhood of v, then

$$J(F) = J(F - v) - J(F - N[v]), \qquad J(\varnothing) = 1, \tag{3}$$

and J is multiplicative over connected components.

Appendix A lists the resulting value for every class. Together with (2) and (3) the table is a self-contained certificate: the recipe of the preceding paragraph fixes a representative of each of the 65 classes, the g-orbits on Q_6 and the graph F_g are read off from that representative, (3) evaluates $J(F_g)$ in integer arithmetic, and (2) supplies the class size. Both sums below are then finite sums of 65 integers, which a reader can recompute directly.

The identity class also checks the one input we take as given. Since 7 and 15 are odd, (1) forces $\Psi(1) = 239 + 14 = 253$, and the value computed here from F_1 by (3) is the entry 253 at type `111111|-` in Appendix A. Thus the reduced Euler characteristic $\widetilde{\chi}(X^{6,4}) = -253$ is recovered from the fixed-orbit data alone, and an error in the Betti numbers of [1, Appendix C.3] would have to leave their alternating sum intact.

Proposition 3. *The class function Ψ satisfies*

$$\sum_{g \in G} \Psi(g) = 138240, \qquad \frac{1}{|G|} \sum_{g \in G} \Psi(g) = 3.$$

Proof. Using (2) and the values in Appendix A, exact summation gives

$$\sum_{\alpha,\beta} |C_{\alpha,\beta}| = 46080 = 2^6 6!, \qquad \sum_{\alpha,\beta} |C_{\alpha,\beta}| \Psi(\alpha; \beta) = 138240.$$

Division gives the stated average. □

Proof of Theorem 1. For a finite-dimensional G-representation V over $\mathbb{Q}$, character averaging gives

$$\frac{1}{|G|}\sum_{g\in G}\operatorname{tr}(g \mid V) = \dim_{\mathbb{Q}} V^G.$$

By (1), the only nonzero reduced homology groups occur in the odd degrees 7 and 15. Moreover $H_i(K;\Bbbk) = \widetilde{H}_i(K;\Bbbk)$ for $i \geq 1$, so the two degrees in question are unaffected by the passage between reduced and unreduced homology. Proposition 3 therefore yields

$$3 = \dim_{\mathbb{Q}} H_7(K;\mathbb{Q})^G + \dim_{\mathbb{Q}} H_{15}(K;\mathbb{Q})^G.$$

At least one summand is at least 2. Since $\mathbb{Q}[G]$ is semisimple, this summand is the multiplicity of the trivial representation.

For any characteristic-zero field $\Bbbk$, extension of scalars gives

$$H_i(K;\Bbbk) \cong H_i(K;\mathbb{Q}) \otimes_{\mathbb{Q}} \Bbbk.$$

Invariants commute with this extension because they are the image of the Reynolds idempotent $|G|^{-1}\sum_{g\in G} g$. Thus the same equality and conclusion hold over $\Bbbk$. $\square$

Appendix A. Conjugacy-class certificate

Partitions are written by concatenating their parts, and `-` denotes the empty partition. Thus `22|11` denotes $((2,2);(1,1))$; the first partition records even-parity cycles and the second odd-parity cycles.

References

[1] F. Galetto, J. Montaño, and Z. Wellner, *Homology of Vietoris–Rips complexes of hypercube graphs via group actions*, arXiv:2606.20784v1, 2026.

Type	$\lvert C\rvert$	Ψ	Type	$\lvert C\rvert$	Ψ	Type	$\lvert C\rvert$	Ψ
`-|111111`	1	1	`-|21111`	30	1	`-|2211`	180	17
`-|222`	120	1	`-|3111`	160	1	`-|321`	960	1
`-|33`	640	1	`-|411`	720	1	`-|42`	1440	1
`-|51`	2304	1	`-|6`	3840	1	`1|11111`	6	1
`1|2111`	120	5	`1|221`	360	17	`1|311`	480	1
`1|32`	960	5	`1|41`	1440	5	`1|5`	2304	1
`11|1111`	15	53	`11|211`	180	5	`11|22`	180	5
`11|31`	480	5	`11|4`	720	5	`111|111`	20	53
`111|21`	120	9	`111|3`	160	5	`1111|11`	15	9
`1111|2`	30	9	`11111|1`	6	9	`111111|-`	1	253
`2|1111`	30	3	`2|211`	360	-1	`2|22`	360	3
`2|31`	960	3	`2|4`	1440	-1	`21|111`	120	3
`21|21`	720	3	`21|3`	960	3	`211|11`	180	7
`211|2`	360	3	`2111|1`	120	7	`21111|-`	30	11
`22|11`	180	9	`22|2`	360	1	`221|1`	360	9
`2211|-`	180	29	`222|-`	120	3	`3|111`	160	-1
`3|21`	960	3	`3|3`	1280	5	`31|11`	480	3
`31|2`	960	3	`311|1`	480	3	`3111|-`	160	7
`32|1`	960	1	`321|-`	960	5	`33|-`	640	13
`4|11`	720	3	`4|2`	1440	3	`41|1`	1440	3
`411|-`	720	7	`42|-`	1440	5	`5|1`	2304	-1
`51|-`	2304	3	`6|-`	3840	3			

TABLE 1. Exact values of $\Psi(g) = I(F_g; -1)$ on all conjugacy classes.